

# ManufactureERP

## Software Requirements Specification

### On-Premise Laravel ERP for Indian Manufacturing SMEs

Version 1.0 | April 2026 | Confidential

|               |                                                   |
|---------------|---------------------------------------------------|
| Document Type | Software Requirements Specification (SRS)         |
| Product       | ManufactureERP — On-Premise Edition               |
| Framework     | Laravel 11, PHP 8.2, MySQL 8, Redis               |
| Target Market | Indian Manufacturing SMEs (Gujarat / Maharashtra) |
| Industries    | Pharma, Chemical, Auto Parts, Plastics, Textiles  |
| Prepared By   | Engineering Team                                  |
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# 1. Introduction & Purpose

ManufactureERP is a comprehensive, on-premise enterprise resource planning system built on Laravel 11. It is designed specifically for small and medium-sized Indian manufacturing businesses in industries such as pharmaceuticals, chemicals, auto parts, plastics, and textiles. The software is deployed directly on the client's own server, ensuring complete data sovereignty, GST compliance, and integration with local business workflows.

## 1.1 Objectives

- Digitize and automate end-to-end manufacturing operations from procurement to dispatch.
- Ensure full GST, e-Invoice (IRN/QR), and e-Way Bill compliance as mandated by the Government of India.
- Provide real-time inventory visibility across multiple warehouses and production floors.
- Reduce manual errors in purchase, sales, and financial transactions.
- Deliver WhatsApp-based instant notifications for critical business events.
- Support role-based access so each department sees only what it needs.
- Remain deployable on client's own VPS or local server without any cloud dependency.

## 1.2 Scope

The system covers 10 core modules: Company Setup, User & Role Management, Vendor Management, Customer Management, Inventory & Warehouse, Purchase Orders, Sales Orders & GST Invoicing, Production & BOM, Finance & Accounts, and HR & Payroll. An optional WhatsApp notification layer ties all modules together for real-time alerts.

## 1.3 Technology Stack

| Layer               | Technology                                       |
|---------------------|--------------------------------------------------|
| Backend Framework   | Laravel 11 (PHP 8.2+)                            |
| Frontend            | Livewire 3 + Alpine.js + Blade Templates         |
| Database            | MySQL 8.0 (with soft deletes on all core tables) |
| Caching & Queues    | Redis + Laravel Horizon                          |
| Job Scheduling      | Laravel Scheduler (cron)                         |
| Notifications       | Laravel Notifications + WhatsApp Cloud API       |
| PDF Generation      | DomPDF (invoices, reports)                       |
| Excel Export        | Laravel Excel (Maatwebsite)                      |
| Roles & Permissions | Spatie Laravel-Permission                        |
| Audit Trail         | Spatie Laravel-Activitylog                       |

| <i>Layer</i>   | <i>Technology</i>                        |
|----------------|------------------------------------------|
| License System | Custom — domain + server fingerprint     |
| Deployment     | Ubuntu 22.04, Nginx, PHP-FPM, Supervisor |

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## 2. How ManufactureERP Works — Overview

ManufactureERP follows a linear yet interconnected business process. Every action in one module triggers downstream effects in other modules. The diagram below illustrates the high-level flow.

### PROCUREMENT CYCLE

Vendor Created → Purchase Requisition → PO Generated → GRN Received → Inventory Updated → Payable Created

### SALES CYCLE

Customer Created → Quotation → Sales Order → Pick & Pack → Invoice (GST) → Dispatch → Payment Received → Receivable Cleared

### PRODUCTION CYCLE

BOM Defined → Production Order → Raw Material Reserved → Production Logged → Finished Goods Added to Inventory

### FINANCE CYCLE

All transactions auto-post journal entries → Ledger Updated → Reports generated (P&L, Balance Sheet, GST Return)

### 2.1 Core Principles

- **Single Source of Truth:** Every stock movement, transaction, and document is recorded once and referenced everywhere.
- **Soft Deletes Everywhere:** No record is permanently deleted. All deletions are logical (deleted\_at timestamp), ensuring full audit trail.
- **Event-Driven Notifications:** Every critical event (low stock, PO approval, overdue payment) fires a WhatsApp message to the responsible person.
- **GST-First Design:** Tax calculation, GSTIN validation, HSN/SAC codes, and e-Invoice generation are built into the core, not added as afterthoughts.
- **Role-Based Isolation:** Users can only access modules and actions permitted by their role. Data is filtered by warehouse/department assignment.
- **Immutable Financial Records:** Posted journal entries cannot be edited — only reversed with a contra entry, maintaining accounting integrity.

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## 3. User Roles & Stories

| Role                  | Responsibilities                                                            |
|-----------------------|-----------------------------------------------------------------------------|
| Super Admin           | Full system access. Manages company settings, license, all modules.         |
| Admin                 | Manages users, masters, and all operational modules. Cannot change license. |
| Purchase Manager      | Manages vendors, purchase requisitions, POs, GRN.                           |
| Sales Manager         | Manages customers, quotations, sales orders, invoices.                      |
| Warehouse Manager     | Manages stock, warehouse transfers, GRN receipt, dispatch.                  |
| Accountant            | Manages accounts payable/receivable, ledgers, GST returns.                  |
| HR Manager            | Manages employees, attendance, leave, payroll.                              |
| Production Supervisor | Manages BOM, production orders, material consumption.                       |
| Staff / Operator      | Read-only or limited data entry within assigned module.                     |
| Vendor (Portal)       | External login to view POs, upload invoices, update delivery status.        |

### 3.1 User Stories

#### US-01 | Purchase Manager

I want to create a Purchase Order with multiple line items, attach a vendor quotation PDF, and send it for approval so that the Finance Manager can review and approve before it is dispatched to the vendor.

#### US-02 | Purchase Manager

I want to receive a WhatsApp alert when any PO I raised is approved or rejected, so I can act immediately without checking the system.

#### US-03 | Warehouse Manager

I want to receive a GRN (Goods Receipt Note) against a PO, inspect quality, accept/partially reject, and have the inventory automatically updated, so stock levels are always accurate.

#### US-04 | Warehouse Manager

I want to get a WhatsApp notification when any item's stock falls below its reorder level, so I can initiate a purchase requisition immediately.

#### US-05 | Sales Manager

I want to convert a Quotation to a Sales Order with one click, auto-calculate GST based on the customer's state (IGST vs CGST+SGST), and generate a GST-compliant invoice PDF that can be emailed and WhatsApp-shared.

### **US-06 | Accountant**

I want every invoice and payment transaction to automatically create the corresponding journal entry so I do not have to do manual double-entry bookkeeping.

### **US-07 | Accountant**

I want to generate GSTR-1 and GSTR-3B data export with a single click at month-end so I can file returns without manual compilation.

### **US-08 | Production Supervisor**

I want to create a Production Order, see the BOM, verify raw material availability, reserve stock, log actual production output, and have finished goods inventory updated automatically.

### **US-09 | HR Manager**

I want to mark employee attendance daily, approve leave applications, and run monthly payroll that auto-calculates PF, ESI, and professional tax deductions.

### **US-10 | Super Admin**

I want to see a real-time dashboard with sales today, pending POs, low-stock items, overdue receivables, and production orders in progress, so I have a complete business snapshot at a glance.

### **US-11 | Vendor (Portal)**

I want to log in to a secure portal, view all POs issued to me, accept or reject them, upload my invoice against a PO, and check payment status without calling the company.

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## 4. Complete Software Flow

### 4.1 Procurement Flow (Step-by-Step)

|         |                                 |                                                                                                                                         |
|---------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Step 1  | <b>Purchase Requisition</b>     | Staff raises PR specifying item, quantity, required date, warehouse. PR status = DRAFT.                                                 |
| Step 2  | <b>PR Approval</b>              | Purchase Manager reviews. Approves → status = APPROVED. Rejects → status = REJECTED with reason. WhatsApp sent to requester.            |
| Step 3  | <b>Vendor Selection / RFQ</b>   | Purchase Manager selects vendor(s). Optional: send RFQ. Vendor responds via portal with quote.                                          |
| Step 4  | <b>Purchase Order Created</b>   | PO created from approved PR. Line items, unit price, GST rate, delivery date entered. PO status = DRAFT.                                |
| Step 5  | <b>PO Approval Workflow</b>     | If PO value > threshold (configurable), requires Finance/Admin approval. Multi-level approval supported. WhatsApp sent at each level.   |
| Step 6  | <b>PO Dispatched to Vendor</b>  | Approved PO emailed + WhatsApp sent to vendor. Vendor can accept/reject via portal. PO status = SENT.                                   |
| Step 7  | <b>GRN (Goods Receipt)</b>      | Warehouse staff receives goods. GRN created against PO. Items inspected. Accepted qty, rejected qty logged. QC photos attachable.       |
| Step 8  | <b>Inventory Update</b>         | GRN posted → inventory levels increased in selected warehouse. Batch/serial numbers assigned if applicable. Stock ledger entry created. |
| Step 9  | <b>Vendor Invoice Matching</b>  | Vendor's invoice matched against GRN. 3-way match: PO qty = GRN qty = Invoice qty. Discrepancies flagged.                               |
| Step 10 | <b>Accounts Payable Created</b> | Matched invoice creates AP entry. Payment scheduled as per vendor credit terms. Journal entry auto-posted.                              |
| Step 11 | <b>Payment Processing</b>       | Accountant records payment (NEFT/RTGS/Cheque). AP cleared. Bank ledger updated. WhatsApp payment confirmation sent to vendor.           |

### 4.2 Sales Flow (Step-by-Step)

|        |                                |                                                                         |
|--------|--------------------------------|-------------------------------------------------------------------------|
| Step 1 | <b>Customer Enquiry / Lead</b> | Sales staff logs enquiry. Customer master created or existing selected. |
|--------|--------------------------------|-------------------------------------------------------------------------|

|         |                             |                                                                                                                                    |
|---------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Step 2  | <b>Quotation</b>            | Items, quantities, prices, GST rates entered. Quotation PDF generated. Sent via email/WhatsApp. Status = SENT.                     |
| Step 3  | <b>Quotation Acceptance</b> | Customer accepts. Quotation converted to Sales Order with one click. Status = CONFIRMED.                                           |
| Step 4  | <b>Credit Limit Check</b>   | System checks if customer's outstanding + new order exceeds credit limit. If yes, requires Admin override.                         |
| Step 5  | <b>Stock Reservation</b>    | Sales Order created → inventory reserved (soft reservation) so same stock isn't sold twice.                                        |
| Step 6  | <b>Pick &amp; Pack</b>      | Warehouse staff picks items per Sales Order. Packaging details recorded. Dispatch challan generated.                               |
| Step 7  | <b>e-Way Bill</b>           | For inter-state or goods value > Rs 50,000, e-Way Bill generated via NIC API. EWB number stored.                                   |
| Step 8  | <b>Invoice Generation</b>   | GST invoice generated with IRN via IRP (Invoice Registration Portal) for e-Invoice. QR code embedded. PDF downloadable.            |
| Step 9  | <b>Dispatch</b>             | Goods dispatched. Courier/vehicle details logged. Inventory deducted. Status = DISPATCHED. WhatsApp with invoice sent to customer. |
| Step 10 | <b>Accounts Receivable</b>  | Invoice creates AR entry. Due date calculated per customer payment terms. Aging tracked.                                           |
| Step 11 | <b>Payment Collection</b>   | Payment received → AR cleared. Receipt generated. Bank ledger updated. WhatsApp receipt sent to customer.                          |

# 5. Module 1 — Company & System Setup

The Company Setup module is the foundation. It is completed once during initial deployment and defines global settings used across all modules.

## 5.1 Company Master Fields

| Field                | Data Type         | Required | Validation Rule                                                   |
|----------------------|-------------------|----------|-------------------------------------------------------------------|
| company_name         | VARCHAR(255)<br>) | Required | Min 3 chars, Max 255 chars                                        |
| legal_name           | VARCHAR(255)<br>) | Required | Legal entity name                                                 |
| gst_in               | CHAR(15)          | Required | Regex: ^[0-9]{2}[A-Z]{5}[0-9]{4}[A-Z]{1}[1-9A-Z]{1}Z[0-9A-Z]{1}\$ |
| pan                  | CHAR(10)          | Required | Regex: ^[A-Z]{5}[0-9]{4}[A-Z]{1}\$                                |
| address_line1        | VARCHAR(255)<br>) | Required | Min 5 chars                                                       |
| address_line2        | VARCHAR(255)<br>) | Optional | —                                                                 |
| city                 | VARCHAR(100)<br>) | Required | Min 2 chars                                                       |
| state_code           | CHAR(2)           | Required | GST state code (e.g. 24 for Gujarat)                              |
| pincode              | CHAR(6)           | Required | Numeric, exactly 6 digits                                         |
| phone                | VARCHAR(15)       | Required | 10-digit Indian mobile or landline                                |
| email                | VARCHAR(255)<br>) | Required | Valid email format                                                |
| logo                 | VARCHAR(500)<br>) | Optional | Image file: JPG/PNG, max 2MB                                      |
| financial_year_start | DATE              | Required | Usually April 1                                                   |
| currency             | CHAR(3)           | Required | Default: INR                                                      |
| invoice_prefix       | VARCHAR(20)       | Required | e.g. INV, GST/2024-25                                             |
| po_prefix            | VARCHAR(20)       | Required | e.g. PO/2024-25                                                   |
| default_tax_type     | ENUM              | Required | IGST or CGST_SGST                                                 |
| whatsapp_enabled     | BOOLEAN           | Required | Default: false                                                    |

| <i>Field</i>                  | <i>Data Type</i> | <i>Required</i> | <i>Validation Rule</i>                     |
|-------------------------------|------------------|-----------------|--------------------------------------------|
| <code>einvoice_enabled</code> | BOOLEAN          | Required        | Default: false (enable for turnover > 5Cr) |

## 6. Module 2 — User & Role Management

Built on Spatie Laravel-Permission. Every user is assigned one or more roles. Roles have fine-grained permissions (e.g. purchase.po.create, inventory.grn.approve). All permission checks use Laravel Gates.

### 6.1 Users Table Fields

| Field           | Type            | Validation / Notes                              |
|-----------------|-----------------|-------------------------------------------------|
| id              | BIGINT UNSIGNED | PK, Auto-increment                              |
| name            | VARCHAR(255)    | Required   Min 2, Max 100 chars                 |
| email           | VARCHAR(255)    | Required   Valid email   Unique                 |
| password        | VARCHAR(255)    | Required   Min 8 chars   Bcrypt hashed          |
| phone           | VARCHAR(15)     | Required   10-digit mobile                      |
| whatsapp_number | VARCHAR(15)     | Optional   10-digit; used for WA notifications  |
| employee_id     | FK → employees  | Nullable   Links user to HR record              |
| warehouse_id    | FK → warehouses | Nullable   Restricts user to specific warehouse |
| is_active       | BOOLEAN         | Default: true   false = login blocked           |
| last_login_at   | TIMESTAMP       | Auto-updated on login                           |
| deleted_at      | TIMESTAMP       | Soft delete   NULL = active                     |
| created_at      | TIMESTAMP       | Auto                                            |
| updated_at      | TIMESTAMP       | Auto                                            |

### 6.2 Default Permission Groups

| Module        | Permissions                                                                      |
|---------------|----------------------------------------------------------------------------------|
| Company Setup | company.view, company.edit                                                       |
| Users         | users.view, users.create, users.edit, users.delete                               |
| Vendors       | vendors.view, vendors.create, vendors.edit, vendors.delete, vendors.approve      |
| Customers     | customers.view, customers.create, customers.edit, customers.delete               |
| Inventory     | inventory.view, inventory.adjust, inventory.transfer, inventory.grn              |
| Purchase      | purchase.pr.create, purchase.po.create, purchase.po.approve, purchase.grn.create |

| Module       | Permissions                                                                         |
|--------------|-------------------------------------------------------------------------------------|
| Sales        | sales.quotation.create, sales.order.create,<br>sales.invoice.create, sales.dispatch |
| Production   | production.bom.create, production.order.create, production.log                      |
| Finance      | finance.voucher.create, finance.payment.approve,<br>finance.reports.view            |
| HR & Payroll | hr.employee.manage, hr.attendance.mark, hr.payroll.run                              |
| Reports      | reports.sales, reports.purchase, reports.inventory,<br>reports.finance              |

## 7. Module 3 — Vendor Management

Vendors are the suppliers of raw materials, packaging, and services. This module manages their master data, documents, credit terms, and performance. Vendors can optionally self-register via the Vendor Portal.

### 7.1 Vendors Table Fields

| Field              | Type            | Validation / Notes                                                 |
|--------------------|-----------------|--------------------------------------------------------------------|
| id                 | BIGINT UNSIGNED | PK                                                                 |
| vendor_code        | VARCHAR(20)     | Required   Auto-generated (VND-0001)   Unique                      |
| company_name       | VARCHAR(255)    | Required   Min 3 chars                                             |
| contact_person     | VARCHAR(100)    | Required   Min 2 chars                                             |
| email              | VARCHAR(255)    | Required   Valid email   Unique per company                        |
| phone              | VARCHAR(15)     | Required   10-digit                                                |
| whatsapp           | VARCHAR(15)     | Optional   10-digit                                                |
| gstln              | CHAR(15)        | Conditional   Required if GST registered   Regex validated         |
| pan                | CHAR(10)        | Required   Regex: <code>^[A-Z]{5}[0-9]{4}[A-Z]{1}\$</code>         |
| address_line1      | VARCHAR(255)    | Required                                                           |
| city               | VARCHAR(100)    | Required                                                           |
| state_code         | CHAR(2)         | Required   GST state code                                          |
| pincode            | CHAR(6)         | Required   Numeric, 6 digits                                       |
| vendor_type        | ENUM            | Required   Values: SUPPLIER, SERVICE, TRANSPORTER, CONTRACTOR      |
| payment_terms_days | SMALLINT        | Required   0–365   e.g. 30 = Net 30                                |
| credit_limit       | DECIMAL(15,2)   | Optional   Min 0                                                   |
| bank_name          | VARCHAR(100)    | Optional                                                           |
| bank_account_no    | VARCHAR(20)     | Optional   Numeric                                                 |
| bank_ifsc          | CHAR(11)        | Optional   Regex: <code>^[A-Z]{4}0[A-Z0-9]{6}\$</code>             |
| status             | ENUM            | Required   Values: ACTIVE, INACTIVE, BLACKLISTED   Default: ACTIVE |
| rating             | DECIMAL(3,2)    | System-calculated   0.00–5.00                                      |
| portal_enabled     | BOOLEAN         | Default: false   Enables vendor portal login                       |

| Field           | Type         | Validation / Notes                                |
|-----------------|--------------|---------------------------------------------------|
| portal_password | VARCHAR(255) | Nullable   Bcrypt hashed   Only if portal_enabled |
| deleted_at      | TIMESTAMP    | Soft delete                                       |

## 7.2 Module Communication

### Vendor Module Triggers

- Purchase Order Module: vendor\_id FK used in all POs. Vendor payment terms auto-fill PO terms.
- Accounts Payable: Vendor bank details used for payment. AP aging linked to vendor.
- Inventory GRN: Vendor linked to GRN for 3-way match.
- WhatsApp: PO sent, payment made, and portal credentials shared via WhatsApp.
- Vendor Portal: Approved vendors with portal\_enabled=true get login credentials.
- Performance Rating: Auto-calculated from on-time delivery % and quality rejection %.

## 8. Module 4 — Customer Management

Customer Management stores B2B client data, credit limits, pricing tiers, and outstanding balances. Every Sales Order, Invoice, and payment links to a customer record.

### 8.1 Customers Table Fields

| Field              | Type             | Validation / Notes                                       |
|--------------------|------------------|----------------------------------------------------------|
| id                 | BIGINT UNSIGNED  | PK                                                       |
| customer_code      | VARCHAR(20)      | Required   Auto: CUS-0001   Unique                       |
| company_name       | VARCHAR(255)     | Required   Min 3 chars                                   |
| contact_person     | VARCHAR(100)     | Required                                                 |
| email              | VARCHAR(255)     | Required   Valid email                                   |
| phone              | VARCHAR(15)      | Required   10-digit                                      |
| whatsapp           | VARCHAR(15)      | Optional   10-digit                                      |
| gst_in             | CHAR(15)         | Conditional   Required for B2B GST invoicing             |
| pan                | CHAR(10)         | Optional                                                 |
| billing_address    | TEXT             | Required                                                 |
| billing_city       | VARCHAR(100)     | Required                                                 |
| billing_state_code | CHAR(2)          | Required   Determines IGST vs CGST+SGST                  |
| billing_pincode    | CHAR(6)          | Required   6 digits                                      |
| shipping_address   | TEXT             | Optional   If blank, same as billing                     |
| customer_type      | ENUM             | Required   Values: DEALER, DISTRIBUTOR, RETAILER, DIRECT |
| price_list_id      | FK → price_lists | Nullable   Customer-specific pricing tier                |
| payment_terms_days | SMALLINT         | Required   0–365                                         |
| credit_limit       | DECIMAL(15,2)    | Required   Default: 0 (unlimited if 0)                   |
| credit_used        | DECIMAL(15,2)    | System-calculated   Sum of unpaid invoices               |
| status             | ENUM             | Required   ACTIVE / INACTIVE / BLOCKED                   |
| deleted_at         | TIMESTAMP        | Soft delete                                              |

### 8.2 GST Tax Logic

GST tax type is determined at invoice time by comparing the seller's state code (from Company Setup) with the buyer's billing\_state\_code:

| Condition                   | Supply Type        | Tax Applied                            |
|-----------------------------|--------------------|----------------------------------------|
| Seller State == Buyer State | Intra-state supply | CGST (half) + SGST (half)              |
| Seller State != Buyer State | Inter-state supply | IGST (full rate)                       |
| Buyer has no GSTIN          | B2C supply         | CGST + SGST or IGST depending on state |
| Export                      | Zero-rated supply  | IGST @ 0% or LUT/Bond                  |

## 9. Module 5 — Inventory & Warehouse

The Inventory module is the central nervous system of ManufactureERP. Every stock movement — purchase receipt, sales dispatch, production consumption, inter-warehouse transfer, and manual adjustment — flows through this module and creates an immutable stock ledger entry.

### 9.1 Products / Items Master

| Field             | Type            | Validation / Notes                                                         |
|-------------------|-----------------|----------------------------------------------------------------------------|
| id                | BIGINT UNSIGNED | PK                                                                         |
| item_code         | VARCHAR(30)     | Required   Auto or manual   Unique                                         |
| item_name         | VARCHAR(255)    | Required   Min 3 chars                                                     |
| description       | TEXT            | Optional                                                                   |
| category_id       | FK → categories | Required                                                                   |
| uom_id            | FK → units      | Required   Unit of Measure (Kg, Pcs, Ltr, etc.)                            |
| hsn_code          | VARCHAR(8)      | Required for GST   4–8 digit numeric                                       |
| gst_rate          | DECIMAL(5,2)    | Required   Values: 0, 5, 12, 18, 28                                        |
| item_type         | ENUM            | Required   RAW_MATERIAL, FINISHED_GOOD, SEMI_FINISHED, CONSUMABLE, SERVICE |
| reorder_level     | DECIMAL(15,3)   | Required   Min 0   Triggers low-stock alert                                |
| reorder_qty       | DECIMAL(15,3)   | Required   Min 0   Suggested PO qty                                        |
| min_stock         | DECIMAL(15,3)   | Optional   Safety stock                                                    |
| max_stock         | DECIMAL(15,3)   | Optional                                                                   |
| purchase_price    | DECIMAL(15,2)   | Optional   Default purchase cost                                           |
| sale_price        | DECIMAL(15,2)   | Optional   Default selling price                                           |
| is_batch_tracked  | BOOLEAN         | Default: false   Enable batch/lot tracking                                 |
| is_serial_tracked | BOOLEAN         | Default: false   Enable serial number tracking                             |
| is_expiry_tracked | BOOLEAN         | Default: false   Relevant for pharma/food                                  |
| barcode           | VARCHAR(50)     | Optional   EAN/QR code                                                     |
| image             | VARCHAR(500)    | Optional   Product image                                                   |
| deleted_at        | TIMESTAMP       | Soft delete                                                                |

## 9.2 Warehouses Table

| Field      | Type            | Notes                            |
|------------|-----------------|----------------------------------|
| id         | BIGINT UNSIGNED | PK                               |
| code       | VARCHAR(20)     | Required   Unique   e.g. WH-MAIN |
| name       | VARCHAR(100)    | Required                         |
| address    | TEXT            | Required                         |
| city       | VARCHAR(100)    | Required                         |
| manager_id | FK → users      | Required                         |
| is_active  | BOOLEAN         | Default: true                    |
| deleted_at | TIMESTAMP       | Soft delete                      |

## 9.3 Stock Ledger (Immutable Movement Log)

Every stock in/out creates one ledger entry. The current stock is always computed as SUM of ledger entries. No direct stock editing is allowed — only adjustment vouchers.

| Field            | Type            | Notes                                                                                            |
|------------------|-----------------|--------------------------------------------------------------------------------------------------|
| id               | BIGINT UNSIGNED | PK                                                                                               |
| item_id          | FK → items      | Required                                                                                         |
| warehouse_id     | FK → warehouses | Required                                                                                         |
| batch_no         | VARCHAR(50)     | Nullable                                                                                         |
| serial_no        | VARCHAR(50)     | Nullable                                                                                         |
| expiry_date      | DATE            | Nullable                                                                                         |
| transaction_type | ENUM            | GRN_IN, SALES_OUT, TRANSFER_IN, TRANSFER_OUT, PRODUCTION_IN, PRODUCTION_OUT, ADJUSTMENT, OPENING |
| reference_type   | VARCHAR(50)     | e.g. App\Models\PurchaseOrder                                                                    |
| reference_id     | BIGINT          | ID of source document                                                                            |
| qty_in           | DECIMAL(15,3)   | Nullable   Stock added                                                                           |
| qty_out          | DECIMAL(15,3)   | Nullable   Stock removed                                                                         |
| balance_qty      | DECIMAL(15,3)   | Running balance at time of entry                                                                 |
| unit_cost        | DECIMAL(15,4)   | Cost per unit at time of transaction                                                             |
| created_by       | FK → users      | Required                                                                                         |
| created_at       | TIMESTAMP       | Auto   NO updated_at — immutable                                                                 |

■ *Stock ledger has NO soft delete and NO updated\_at. It is append-only. To reverse a wrong entry, create an opposite adjustment entry.*

# 10. Module 6 — Purchase Order Management

The Purchase module manages the full procurement lifecycle: Purchase Requisition → Purchase Order → Goods Receipt Note → Vendor Invoice → Accounts Payable.

## 10.1 Purchase Requisitions Table

| Field           | Type             | Notes                                                  |
|-----------------|------------------|--------------------------------------------------------|
| id              | BIGINT UNSIGNED  | PK                                                     |
| pr_number       | VARCHAR(30)      | Auto-generated   PR/2024-25/0001                       |
| requested_by    | FK → users       | Required                                               |
| department_id   | FK → departments | Required                                               |
| required_date   | DATE             | Required   Must be >= today                            |
| notes           | TEXT             | Optional   Max 1000 chars                              |
| status          | ENUM             | DRAFT, PENDING_APPROVAL, APPROVED, REJECTED, CONVERTED |
| approved_by     | FK → users       | Nullable   Set when approved                           |
| approved_at     | TIMESTAMP        | Nullable                                               |
| rejected_reason | TEXT             | Nullable   Required if REJECTED                        |
| deleted_at      | TIMESTAMP        | Soft delete                                            |

## 10.2 Purchase Requisition Items Table

| Field           | Type                       | Notes                              |
|-----------------|----------------------------|------------------------------------|
| id              | BIGINT UNSIGNED            | PK                                 |
| pr_id           | FK → purchase_requisitions | Required                           |
| item_id         | FK → items                 | Required                           |
| qty_requested   | DECIMAL(15,3)              | Required   Min: 0.001              |
| uom_id          | FK → units                 | Required                           |
| estimated_price | DECIMAL(15,2)              | Optional                           |
| purpose         | VARCHAR(255)               | Optional   Why is this item needed |

## 10.3 Purchase Orders Table

| Field              | Type                       | Notes                                                                 |
|--------------------|----------------------------|-----------------------------------------------------------------------|
| id                 | BIGINT UNSIGNED            | PK                                                                    |
| po_number          | VARCHAR(30)                | Auto   PO/2024-25/0001   Unique                                       |
| pr_id              | FK → purchase_requisitions | Nullable   Source PR                                                  |
| vendor_id          | FK → vendors               | Required                                                              |
| warehouse_id       | FK → warehouses            | Required   Delivery location                                          |
| order_date         | DATE                       | Required   Default: today                                             |
| expected_delivery  | DATE                       | Required   Must be >= order_date                                      |
| payment_terms_days | SMALLINT                   | Required   Auto-filled from vendor                                    |
| subtotal           | DECIMAL(15,2)              | System-calculated                                                     |
| discount_amount    | DECIMAL(15,2)              | Optional   Min 0                                                      |
| taxable_amount     | DECIMAL(15,2)              | System-calculated                                                     |
| cgst_amount        | DECIMAL(15,2)              | System-calculated                                                     |
| sgst_amount        | DECIMAL(15,2)              | System-calculated                                                     |
| igst_amount        | DECIMAL(15,2)              | System-calculated                                                     |
| total_amount       | DECIMAL(15,2)              | System-calculated                                                     |
| status             | ENUM                       | DRAFT, PENDING_APPROVAL, APPROVED, SENT, PARTIAL, RECEIVED, CANCELLED |
| approval_level     | TINYINT                    | 0–3   Configurable by amount threshold                                |
| notes              | TEXT                       | Optional                                                              |
| deleted_at         | TIMESTAMP                  | Soft delete                                                           |

## 10.4 GRN (Goods Receipt Note) Table

| Field         | Type                 | Notes                     |
|---------------|----------------------|---------------------------|
| id            | BIGINT UNSIGNED      | PK                        |
| grn_number    | VARCHAR(30)          | Auto   GRN/2024-25/0001   |
| po_id         | FK → purchase_orders | Required                  |
| vendor_id     | FK → vendors         | Required   Auto from PO   |
| warehouse_id  | FK → warehouses      | Required                  |
| received_date | DATE                 | Required   Default: today |

| Field      | Type        | Notes                           |
|------------|-------------|---------------------------------|
| vehicle_no | VARCHAR(20) | Optional                        |
| lr_number  | VARCHAR(50) | Optional   Lorry Receipt number |
| status     | ENUM        | DRAFT, POSTED, CANCELLED        |
| notes      | TEXT        | Optional                        |
| deleted_at | TIMESTAMP   | Soft delete                     |

■ Posting a GRN is irreversible. It creates stock ledger entries and an AP invoice. To reverse, a GRN Return must be created.

# 11. Module 7 — Sales Order & GST Invoicing

The Sales module handles the complete order-to-cash cycle. It is tightly integrated with Inventory (stock reservation and deduction), Finance (AR creation), and the GST e-Invoice system.

## 11.1 Quotations Table

| Field            | Type            | Notes                                               |
|------------------|-----------------|-----------------------------------------------------|
| id               | BIGINT UNSIGNED | PK                                                  |
| quotation_number | VARCHAR(30)     | Auto   QT/2024-25/0001                              |
| customer_id      | FK → customers  | Required                                            |
| quotation_date   | DATE            | Required   Default: today                           |
| valid_until      | DATE            | Required   Must be > quotation_date                 |
| subtotal         | DECIMAL(15,2)   | System-calculated                                   |
| discount_amount  | DECIMAL(15,2)   | Optional                                            |
| total_amount     | DECIMAL(15,2)   | System-calculated                                   |
| status           | ENUM            | DRAFT, SENT, ACCEPTED, REJECTED, EXPIRED, CONVERTED |
| notes            | TEXT            | Optional   Printed on quotation PDF                 |
| terms_conditions | TEXT            | Optional                                            |
| deleted_at       | TIMESTAMP       | Soft delete                                         |

## 11.2 Sales Orders Table

| Field              | Type            | Notes                                   |
|--------------------|-----------------|-----------------------------------------|
| id                 | BIGINT UNSIGNED | PK                                      |
| so_number          | VARCHAR(30)     | Auto   SO/2024-25/0001                  |
| quotation_id       | FK → quotations | Nullable   If converted from quotation  |
| customer_id        | FK → customers  | Required                                |
| warehouse_id       | FK → warehouses | Required   Dispatch from                |
| order_date         | DATE            | Required                                |
| expected_dispatch  | DATE            | Required   Must be >= order_date        |
| payment_terms_days | SMALLINT        | Required   Auto from customer           |
| billing_address    | TEXT            | Required   Auto from customer, editable |

| Field            | Type          | Notes                                                         |
|------------------|---------------|---------------------------------------------------------------|
| shipping_address | TEXT          | Required   Auto from customer, editable                       |
| subtotal         | DECIMAL(15,2) | System-calculated                                             |
| discount_amount  | DECIMAL(15,2) | Optional                                                      |
| taxable_amount   | DECIMAL(15,2) | System-calculated                                             |
| cgst_amount      | DECIMAL(15,2) | System-calculated                                             |
| sgst_amount      | DECIMAL(15,2) | System-calculated                                             |
| igst_amount      | DECIMAL(15,2) | System-calculated                                             |
| total_amount     | DECIMAL(15,2) | System-calculated                                             |
| status           | ENUM          | DRAFT, CONFIRMED, PROCESSING, DISPATCHED, INVOICED, CANCELLED |
| deleted_at       | TIMESTAMP     | Soft delete                                                   |

### 11.3 Invoices (GST) Table

| Field            | Type              | Notes                                                 |
|------------------|-------------------|-------------------------------------------------------|
| id               | BIGINT UNSIGNED   | PK                                                    |
| invoice_number   | VARCHAR(30)       | Auto   Format per company config   Unique             |
| so_id            | FK → sales_orders | Required                                              |
| customer_id      | FK → customers    | Required                                              |
| invoice_date     | DATE              | Required                                              |
| due_date         | DATE              | System-calculated   invoice_date + payment_terms_days |
| place_of_supply  | CHAR(2)           | Required   Customer billing state code                |
| reverse_charge   | BOOLEAN           | Default: false                                        |
| irn              | VARCHAR(64)       | Nullable   e-Invoice IRN (from IRP)                   |
| irn_generated_at | TIMESTAMP         | Nullable                                              |
| qr_code_image    | TEXT              | Nullable   Base64 QR from IRP                         |
| eway_bill_number | VARCHAR(12)       | Nullable                                              |
| subtotal         | DECIMAL(15,2)     | System-calculated                                     |
| discount_amount  | DECIMAL(15,2)     | Optional                                              |
| taxable_amount   | DECIMAL(15,2)     | System-calculated                                     |
| cgst_amount      | DECIMAL(15,2)     | System-calculated                                     |

| Field            | Type          | Notes                                          |
|------------------|---------------|------------------------------------------------|
| sgst_amount      | DECIMAL(15,2) | System-calculated                              |
| igst_amount      | DECIMAL(15,2) | System-calculated                              |
| tcs_amount       | DECIMAL(15,2) | Optional   TCS if applicable                   |
| total_amount     | DECIMAL(15,2) | System-calculated                              |
| amount_in_words  | VARCHAR(500)  | Auto-generated                                 |
| status           | ENUM          | DRAFT, POSTED, PARTIALLY_PAID, PAID, CANCELLED |
| cancelled_reason | TEXT          | Nullable   Required if CANCELLED               |
| deleted_at       | TIMESTAMP     | Soft delete                                    |

## 12. Module 8 — Production & Bill of Materials

The Production module enables manufacturers to define what raw materials and semi-finished goods go into making each finished product (BOM), create production orders, track work-in-progress, log actual production, and automatically update inventory.

### 12.1 Bill of Materials (BOM)

| Field        | Type            | Notes                                                    |
|--------------|-----------------|----------------------------------------------------------|
| id           | BIGINT UNSIGNED | PK                                                       |
| bom_code     | VARCHAR(30)     | Auto   BOM-0001                                          |
| product_id   | FK → items      | Required   Finished good being produced                  |
| version      | VARCHAR(10)     | Required   e.g. v1.0   Multiple BOM versions allowed     |
| qty_produced | DECIMAL(15,3)   | Required   How many units this BOM produces   Min: 0.001 |
| uom_id       | FK → units      | Required                                                 |
| is_active    | BOOLEAN         | Default: true   Only one active BOM per product          |
| notes        | TEXT            | Optional                                                 |
| deleted_at   | TIMESTAMP       | Soft delete                                              |

### 12.2 Production Orders Table

| Field             | Type            | Notes                               |
|-------------------|-----------------|-------------------------------------|
| id                | BIGINT UNSIGNED | PK                                  |
| prod_order_number | VARCHAR(30)     | Auto   PRD/2024-25/0001             |
| bom_id            | FK → boms       | Required                            |
| product_id        | FK → items      | Required   Auto from BOM            |
| warehouse_id      | FK → warehouses | Required   Where finished goods go  |
| planned_qty       | DECIMAL(15,3)   | Required   Min: 0.001               |
| actual_qty        | DECIMAL(15,3)   | Nullable   Filled on completion     |
| planned_start     | DATE            | Required                            |
| planned_end       | DATE            | Required   Must be >= planned_start |
| actual_start      | TIMESTAMP       | Nullable                            |
| actual_end        | TIMESTAMP       | Nullable                            |

| Field      | Type      | Notes                                              |
|------------|-----------|----------------------------------------------------|
| status     | ENUM      | DRAFT, RELEASED, IN_PROGRESS, COMPLETED, CANCELLED |
| deleted_at | TIMESTAMP | Soft delete                                        |

### Production Module Communication

- Inventory Module: On RELEASE, raw materials are soft-reserved. On COMPLETION, raw materials are deducted (PRODUCTION\_OUT) and finished goods added (PRODUCTION\_IN) via stock ledger.
- BOM Module: Planned material quantities = BOM component qty × (planned\_qty / bom.qty\_produced).
- Sales Order Module: Production orders can be auto-suggested when a Sales Order is confirmed but stock is insufficient.
- WhatsApp: Notification sent to Production Supervisor when order is released, and to Admin when completed.

---

## 13. Module 9 — Finance & Accounts

Finance is the read-only aggregation layer of ManufactureERP. Journal entries are auto-posted by other modules (Purchase, Sales, Payments). The Accountant's role is to review, reconcile, and report — not to re-enter data that is already captured.

### 13.1 Chart of Accounts

| Field        | Type            | Notes                                     |
|--------------|-----------------|-------------------------------------------|
| id           | BIGINT UNSIGNED | PK                                        |
| account_code | VARCHAR(20)     | Required   Unique   e.g. 1001, 2001       |
| account_name | VARCHAR(255)    | Required                                  |
| account_type | ENUM            | ASSET, LIABILITY, EQUITY, INCOME, EXPENSE |
| parent_id    | FK → accounts   | Nullable   For hierarchical COA           |
| is_system    | BOOLEAN         | System accounts cannot be deleted         |
| deleted_at   | TIMESTAMP       | Soft delete (non-system only)             |

### 13.2 Journal Entries — Auto-Posting Rules

| Trigger Event         | Journal Entry                                                 |
|-----------------------|---------------------------------------------------------------|
| GRN Posted            | Dr: Inventory A/c   Cr: Accounts Payable A/c                  |
| Sales Invoice Posted  | Dr: Accounts Receivable A/c   Cr: Sales A/c + GST Payable A/c |
| Payment to Vendor     | Dr: Accounts Payable A/c   Cr: Bank / Cash A/c                |
| Payment from Customer | Dr: Bank / Cash A/c   Cr: Accounts Receivable A/c             |
| Production Completed  | Dr: Finished Goods A/c   Cr: WIP A/c                          |
| Salary Paid           | Dr: Salary Expense A/c   Cr: Bank A/c + Statutory Liabilities |
| Stock Adjustment      | Dr/Cr: Inventory A/c   Cr/Dr: Inventory Adjustment A/c        |

## 14. Module 10 — HR & Payroll

### 14.1 Employees Table

| Field           | Type              | Notes                                        |
|-----------------|-------------------|----------------------------------------------|
| id              | BIGINT UNSIGNED   | PK                                           |
| employee_code   | VARCHAR(20)       | Auto   EMP-0001   Unique                     |
| first_name      | VARCHAR(100)      | Required   Min 2 chars                       |
| last_name       | VARCHAR(100)      | Required                                     |
| date_of_birth   | DATE              | Required   Age must be 18–65                 |
| gender          | ENUM              | MALE, FEMALE, OTHER                          |
| phone           | VARCHAR(15)       | Required   10-digit                          |
| email           | VARCHAR(255)      | Optional   Valid email                       |
| whatsapp        | VARCHAR(15)       | Optional                                     |
| department_id   | FK → departments  | Required                                     |
| designation_id  | FK → designations | Required                                     |
| date_of_joining | DATE              | Required   Cannot be future date             |
| employment_type | ENUM              | FULL_TIME, PART_TIME, CONTRACT, DAILY_WAGE   |
| basic_salary    | DECIMAL(10,2)     | Required   Min 0                             |
| hra             | DECIMAL(10,2)     | Optional                                     |
| pf_number       | VARCHAR(22)       | Optional   PF applicable if salary <= 15000  |
| esi_number      | VARCHAR(17)       | Optional   ESI applicable if salary <= 21000 |
| pan             | CHAR(10)          | Required   Regex validated                   |
| aadhaar         | CHAR(12)          | Optional   12-digit numeric                  |
| bank_account_no | VARCHAR(20)       | Optional                                     |
| bank_ifsc       | CHAR(11)          | Optional   Regex validated                   |
| status          | ENUM              | ACTIVE, RESIGNED, TERMINATED, ON_LEAVE       |
| deleted_at      | TIMESTAMP         | Soft delete                                  |

---

## 15. Module Communication Map

The table below shows which modules emit data, which modules consume it, and the communication mechanism (DB FK, Laravel Event, Queue Job, or WhatsApp Notification).

| <i>Event / Trigger</i> | <i>Receiving Module</i> | <i>Mechanism</i>                               |
|------------------------|-------------------------|------------------------------------------------|
| Vendor Created         | Purchase Orders         | FK: vendor_id   Auto-fill payment terms        |
| Vendor Created         | Accounts Payable        | FK: vendor_id   Payment processing             |
| Customer Created       | Quotations              | FK: customer_id   Auto-fill tax type           |
| Customer Created       | Sales Orders            | FK: customer_id   Credit limit check           |
| PO Approved            | WhatsApp Service        | Event: PurchaseOrderApproved → WA to vendor    |
| GRN Posted             | Inventory               | Stock ledger: PRODUCTION_IN entry created      |
| GRN Posted             | Finance                 | Auto: AP invoice + Journal Entry created       |
| Sales Order Confirmed  | Inventory               | Stock reservation created for all line items   |
| Invoice Posted         | Finance                 | Auto: AR entry + Journal Entry created         |
| Invoice Posted         | WhatsApp Service        | Event: InvoiceCreated → WA + PDF to customer   |
| Stock Below Reorder    | WhatsApp Service        | Scheduled Job → WA to Purchase Manager         |
| Payment Received       | Finance                 | AR cleared, Journal Entry reversed/settled     |
| Production Released    | Inventory               | Raw materials soft-reserved                    |
| Production Complete    | Inventory               | Deduct RM, Add FG via stock ledger             |
| Salary Run             | Finance                 | Payroll Journal Entry auto-posted              |
| License Expired        | All Modules             | Middleware blocks access, shows renewal screen |

## 16. Database Schema & Soft Deletes

All tables (except stock\_ledger, journal\_entry\_lines, and audit\_log) include a deleted\_at TIMESTAMP column for soft deletes. Laravel's SoftDeletes trait is used on all Eloquent models. Soft-deleted records are excluded from all queries by default and can be restored if needed.

### 16.1 Soft Delete Implementation

- All models use: use SoftDeletes; at the top.
- All foreign keys referencing soft-deletable tables must check deleted\_at IS NULL in queries.
- A GlobalScope filters out soft-deleted records in all queries automatically.
- Admin can view deleted records via withTrashed() and restore via restore() method.
- forceDelete() is disabled for financial records (invoices, journal entries, payments) — these can only be cancelled.
- Cascade soft deletes: Deleting a Vendor soft-deletes related POs (only if DRAFT status).

### 16.2 Complete Table List

| Table Name           | Purpose                         | Soft Delete |
|----------------------|---------------------------------|-------------|
| companies            | Company master — single record  | Y           |
| users                | System users                    | Y           |
| roles                | Spatie roles                    | N           |
| permissions          | Spatie permissions              | N           |
| role_has_permissions | Pivot: role-permission          | N           |
| model_has_roles      | Pivot: user-role                | N           |
| vendors              | Supplier/vendor master          | Y           |
| vendor_documents     | Vendor KYC docs                 | Y           |
| customers            | Customer master                 | Y           |
| customer_addresses   | Multiple ship-to addresses      | Y           |
| price_lists          | Customer-specific pricing       | Y           |
| price_list_items     | Price list line items           | Y           |
| categories           | Item categories                 | Y           |
| units_of_measure     | UOM master (Kg, Pcs, Ltr)       | Y           |
| items                | Products / raw materials master | Y           |
| warehouses           | Warehouse / godown master       | Y           |
| stock_ledger         | Immutable stock movement log    | N           |

| Table Name                 | Purpose                             | Soft Delete |
|----------------------------|-------------------------------------|-------------|
| stock_reservations         | Soft reservations for SO/Production | Y           |
| departments                | Company departments                 | Y           |
| purchase_requisitions      | PR header                           | Y           |
| purchase_requisition_items | PR line items                       | Y           |
| purchase_orders            | PO header                           | Y           |
| purchase_order_items       | PO line items                       | Y           |
| po_approvals               | Approval workflow log               | N           |
| goods_receipt_notes        | GRN header                          | Y           |
| grn_items                  | GRN line items                      | Y           |
| quotations                 | Sales quotation header              | Y           |
| quotation_items            | Quotation line items                | Y           |
| sales_orders               | SO header                           | Y           |
| sales_order_items          | SO line items                       | Y           |
| invoices                   | GST invoice header                  | Y           |
| invoice_items              | Invoice line items                  | Y           |
| dispatch_challans          | Delivery challans                   | Y           |
| boms                       | Bill of Materials header            | Y           |
| bom_components             | BOM raw material list               | Y           |
| production_orders          | Production order header             | Y           |
| production_order_materials | Actual material consumption         | N           |
| accounts                   | Chart of accounts                   | Y           |
| journal_entries            | Journal entry header                | N           |
| journal_entry_lines        | Double-entry lines (Dr/Cr)          | N           |
| payments                   | Vendor/customer payments            | Y           |
| employees                  | Employee master                     | Y           |
| attendance                 | Daily attendance log                | N           |
| leave_applications         | Leave requests                      | Y           |
| payroll_runs               | Monthly payroll batch               | Y           |
| payroll_details            | Per-employee payslip                | Y           |

| Table Name    | Purpose                     | Soft Delete |
|---------------|-----------------------------|-------------|
| whatsapp_logs | All WA messages sent/failed | N           |
| license       | License key and expiry      | N           |
| activity_log  | Spatie audit trail          | N           |
| settings      | Key-value system settings   | N           |

---

## 17. WhatsApp API Integration

ManufactureERP integrates with the WhatsApp Cloud API (Meta Business Platform) to send real-time notifications. All messages are queued via Laravel Horizon to prevent blocking the main thread. Message delivery status is tracked in the whatsapp\_logs table.

### 17.1 Setup Requirements

- Meta Business Account with verified business.
- WhatsApp Business API — Phone Number ID and Access Token stored in .env file.
- Message templates pre-approved by Meta for each notification type.
- WABA\_PHONE\_NUMBER\_ID and WABA\_ACCESS\_TOKEN in .env.
- Laravel Horizon running to process the notification queue.

### 17.2 WhatsApp Notification Events

| ID                    | Event             | Triggered By     | Recipient Field           | Details                                                                |
|-----------------------|-------------------|------------------|---------------------------|------------------------------------------------------------------------|
| W<br>A<br>-<br>0<br>1 | PO Approved       | Purchase Manager | vendor_whatsapp           | Template: po_approved   Vars: PO No, Amount, Delivery Date             |
| W<br>A<br>-<br>0<br>2 | PO Sent to Vendor | System           | vendor_whatsapp           | Template: po_dispatch   Vars: PO No, Items summary, Portal link        |
| W<br>A<br>-<br>0<br>3 | Low Stock Alert   | Scheduler        | purchase_manager_whatsapp | Template: low_stock   Vars: Item name, Current qty, Reorder level      |
| W<br>A<br>-<br>0<br>4 | GRN Posted        | Warehouse Mgr    | purchase_manager_whatsapp | Template: grn_posted   Vars: GRN No, Vendor, Items received            |
| W<br>A<br>-<br>0<br>5 | Invoice Generated | Sales Module     | customer_whatsapp         | Template: invoice_sent   Vars: Inv No, Amount, Due date   PDF attached |

| ID    | Event               | Triggered By      | Recipient Field     | Details                                                        |
|-------|---------------------|-------------------|---------------------|----------------------------------------------------------------|
| WA-06 | Payment Overdue     | Scheduler (daily) | customer_whatsapp   | Template: payment_overdue   Vars: Inv No, Days overdue, Amount |
| WA-07 | Payment Received    | Finance           | customer_whatsapp   | Template: payment_receipt   Vars: Receipt No, Amount, Balance  |
| WA-08 | Payment to Vendor   | Finance           | vendor_whatsapp     | Template: payment_sent   Vars: Payment No, Amount, UTR         |
| WA-09 | Production Started  | Production        | supervisor_whatsapp | Template: prod_started   Vars: Order No, Product, Qty          |
| WA-10 | Production Complete | Production        | admin_whatsapp      | Template: prod_complete   Vars: Order No, Actual qty, Variance |
| WA-11 | Leave Approved      | HR                | employee_whatsapp   | Template: leave_approved   Vars: Dates, Type, Approved by      |
| WA-12 | Salary Credited     | Payroll           | employee_whatsapp   | Template: salary_credited   Vars: Month, Net salary            |
| WA-13 | Vendor Portal Creds | Admin             | vendor_whatsapp     | Template: portal_login   Vars: URL, Username, Password         |
| WA-14 | License Expiry      | Scheduler         | admin_whatsapp      | Template: license_expiry   Vars: Days remaining, Renewal URL   |

## 17.3 WhatsApp Service — Laravel Implementation

A dedicated WhatsAppService class handles all message sending. Notifications extend Laravel's Notification class and use a custom WhatsApp channel.

```
// app/Services/WhatsAppService.php

public function sendTemplate(string $to, string $template, array $vars): void {
    dispatch(new SendWhatsAppJob($to, $template, $vars));
}

// app/Jobs/SendWhatsAppJob.php
Http::withToken(config('whatsapp.access_token'))
->post('https://graph.facebook.com/v18.0/{phone_id}/messages', [
    'messaging_product' => 'whatsapp',
    'to' => $this->number,
    'type' => 'template',
    'template' => ['name' => $this->template,
    'language' => ['code' => 'en'],
    'components' => $this->buildComponents()],
]);
```

## 17.4 WhatsApp Logs Table

| Field            | Type            | Notes                                 |
|------------------|-----------------|---------------------------------------|
| id               | BIGINT UNSIGNED | PK                                    |
| event_type       | VARCHAR(50)     | e.g. invoice_sent, low_stock          |
| reference_type   | VARCHAR(100)    | Polymorphic model class               |
| reference_id     | BIGINT          | Source record ID                      |
| recipient_name   | VARCHAR(100)    | Who received it                       |
| recipient_number | VARCHAR(15)     | WhatsApp number sent to               |
| template_name    | VARCHAR(100)    | Meta template used                    |
| message_id       | VARCHAR(100)    | Meta's message ID for tracking        |
| status           | ENUM            | QUEUED, SENT, DELIVERED, READ, FAILED |
| error_message    | TEXT            | Nullable   If failed                  |
| sent_at          | TIMESTAMP       | When dispatched                       |

# 18. License & Deployment System

- On deployment, a unique fingerprint is generated from: domain name + server MAC address + CPU ID.
- This fingerprint is sent to your License Server to obtain a signed license token (JWT).
- The token is stored locally in the database (license table) and verified on every app boot via a ServiceProvider.
- If license is expired or fingerprint changes (server migration), app enters read-only lockout mode.
- AMC renewal: client pays → you issue new token from your admin panel → client runs: php artisan license:refresh
- Module-level licensing: individual modules (e.g. Production, Payroll) can be licensed separately.

| Field              | Type         | Notes                                |
|--------------------|--------------|--------------------------------------|
| id                 | INT          | PK                                   |
| license_key        | VARCHAR(64)  | Unique   Issued by vendor            |
| client_name        | VARCHAR(255) | Client's company name                |
| domain             | VARCHAR(255) | Bound domain (e.g. erp.clientco.com) |
| server_fingerprint | VARCHAR(255) | MAC+CPU hash                         |
| token              | TEXT         | Signed JWT from license server       |
| modules_enabled    | JSON         | Array of enabled module keys         |
| issued_at          | DATE         | License issue date                   |
| expires_at         | DATE         | AMC expiry date                      |
| is_active          | BOOLEAN      | false = locked                       |

## 19. Validation Rules Reference

All input validation is handled by Laravel Form Requests. Each module has its own Request class (e.g. StorePurchaseOrderRequest). The following global rules apply across all modules.

| Field Type    | Regex / Laravel Rule                                                    | Notes                          |
|---------------|-------------------------------------------------------------------------|--------------------------------|
| GSTIN         | <code>^[0-9]{2}[A-Z]{5}[0-9]{4}[A-Z]{1}[1-9A-Z]{1}Z[0-9A-Z]{1}\$</code> | 15 characters, format-verified |
| PAN           | <code>^[A-Z]{5}[0-9]{4}[A-Z]{1}\$</code>                                | 10 chars, uppercase            |
| IFSC          | <code>^[A-Z]{4}0[A-Z0-9]{6}\$</code>                                    | 11 chars                       |
| Pincode       | <code>^[1-9][0-9]{5}\$</code>                                           | 6-digit, cannot start with 0   |
| Mobile        | <code>^[6-9][0-9]{9}\$</code>                                           | 10-digit, starts 6-9           |
| Email         | <code>email:rfc,dns</code>                                              | RFC + DNS verified             |
| HSN Code      | <code>^[0-9]{4,8}\$</code>                                              | 4 to 8 digits                  |
| Amount Fields | <code>numeric min:0 max:99999999.99</code>                              | Positive decimal               |
| Qty Fields    | <code>numeric min:0.001 max:99999999</code>                             | Min 0.001                      |
| Date (future) | <code>date after_or_equal:today</code>                                  | For delivery dates, expiry     |
| File Uploads  | <code>mimes:pdf,jpg,png max:5120</code>                                 | Max 5MB, limited types         |
| Passwords     | <code>min:8 regex(upper+lower+number)</code>                            | Strong password policy         |
| Status ENUM   | <code>in:DRAFT,APPROVED,...</code>                                      | Only defined ENUM values       |

### 19.1 Business Logic Validations

- PO total\_amount must match SUM(line items × qty × rate) — server-side recalculation always.
- GRN accepted\_qty + rejected\_qty must equal received\_qty for each line.
- Invoice taxable\_amount = subtotal - discount\_amount. GST computed from taxable\_amount × rate.
- Credit limit check: if customer.credit\_limit > 0, ensure (credit\_used + new\_order\_total) <= credit\_limit.
- Inventory: qty\_out on stock ledger cannot exceed current available balance for that item+warehouse+batch.
- PO cannot be approved by the same user who created it (4-eye principle).
- BOM: A component item cannot be the same as the product being produced (circular BOM prevention).
- Payroll: PF deduction only if employee basic\_salary <= 15,000. ESI only if gross <= 21,000.
- e-Invoice: IRN generation required for B2B invoices if company turnover > 5 Crore (configurable).

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## 20. Non-Functional Requirements

### Performance

- Page load time < 2 seconds for standard views (list/form) on LAN.
- Report generation < 10 seconds for date ranges up to 1 year.
- PDF invoice generation < 3 seconds.
- Stock balance query < 1 second (optimised via indexed stock\_ledger table).

### Security

- All passwords hashed with Bcrypt (cost factor 12).
- CSRF protection on all forms (Laravel default).
- SQL injection prevention via Eloquent parameterized queries.
- XSS prevention via Blade auto-escaping {{ }}.
- Rate limiting on login: max 5 attempts per minute per IP.
- HTTPS enforced via Nginx config. HTTP → HTTPS redirect.
- Sensitive .env file outside web root. No .env in version control.

### Data Integrity

- Database transactions used for all multi-table operations (GRN posting, Invoice creation).
- Foreign key constraints enforced at DB level.
- Soft deletes prevent orphaned reference data.
- Audit log (Spatie ActivityLog) records every create/update/delete with user and timestamp.

### Backup

- Daily automated DB backup via Laravel Backup package.
- Backup stored locally + optionally on Google Drive / S3.
- Backup retention: 30 days local, 90 days remote.
- Backup failure sends WhatsApp alert to Admin.

### Availability

- System designed for 99.5% uptime on client's local server.
- Supervisor ensures queue workers restart automatically on failure.
- Health check endpoint: /health returns server status (DB, Redis, Queue).

## Scalability

- Single-tenant per deployment — no resource contention between clients.
- MySQL query optimized with composite indexes on high-query tables (stock\_ledger, invoices, journal\_entry\_lines).
- Designed to handle up to 500 concurrent users and 5 years of transaction data without performance degradation.

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